Phase-Modulated Compound Sparsity: Strategy-Dependent Severity Gradients in Reasoning Models

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Abstract

Compound sparsity—simultaneously pruning attention heads and evicting KV cache entries—is essential for efficient LLM inference but produces super-additive quality degradation whose structure is poorly understood. Prior work treats compound interaction magnitude as approximately constant across the generation process. We show that this assumption is wrong: compound sparsity severity is an emergent phase-strategy interaction that varies by over an order of magnitude across reasoning phases.

We evaluate compound sparsity on DeepSeek-R1-Distill-Qwen-7B and Qwen2.5-7B-Instruct (both Qwen2 architecture, 7B parameters) using 50 MATH-500 problems, measuring the Interaction Index—a super-additivity metric—across quantile-segmented reasoning phases at sparsity levels $s \in \{0.15, 0.20, 0.25, 0.30\}$. Under importance-based KV cache eviction, the median Interaction Index difference between low-sparsity and high-sparsity reasoning phases increases monotonically from 0.39 at s=0.15 to 14.59 at s=0.30, with paired Cohen's d_z growing from 0.44 (small) to 0.80 (large); Bonferroni-corrected permutation tests yield $p < 0.005$ at all four sparsity levels. Random eviction produces no significant phase modulation in 7/8 conditions; the single exception (Qwen s=0.30, p=0.012) shows negligible effect size (d_z=−0.04), confirming that the severity gradient is strategy-specific, not an inherent property of compound sparsity. At s=0.25, a crossover emerges: importance-based eviction outperforms random in high-sparsity phases ($\widetilde{\mathrm{II}} = 0.032$ vs. 0.818) yet underperforms in low-sparsity phases ($\widetilde{\mathrm{II}} = 7.196$ vs. 1.363)—a qualitative reversal in strategy ranking. The pattern replicates on Qwen2.5-7B-Instruct at $s \geq 0.20$ ($d_z \geq 0.55$; Spearman $\rho = 1.0$ across sparsity levels; gradient permutation $p = 0.043$).

We propose entropy-discrimination collapse as an explanatory mechanism: head pruning flattens cumulative importance scores in phases where attention is already broadly distributed, degrading the discriminative power of importance-based eviction in a phase-dependent manner. These findings challenge the phase-invariance assumption in compound sparsity deployment and motivate phase-aware compression budgets for reasoning models.

1 Introduction

Long-context language models place acute pressure on the key–value (KV) cache, which grows linearly with sequence length and dominates inference memory at scale (Kwon et al., 2023; Pope et al., 2023). Two complementary responses have emerged: *attention head pruning*, which removes structurally redundant heads (Voita et al., 2019; Michel et al., 2019), and *KV cache eviction*, which discards low-importance token positions from the cache (Li et al., 2024; Feng et al., 2025). Deploying both simultaneously—*compound sparsity*—is attractive because the compression gains multiply, but the resulting quality degradation can be super-additive: the joint loss exceeds the sum of individual losses (Liao et al., 2026; Mozaffari et al., 2025).

Prior work has characterized compound interaction as a property of the sparsity level—higher sparsity yields more interaction—but treats the interaction magnitude as roughly constant across the generation process (Liao et al., 2026; von Rad et al., 2026). This assumption is problematic for reasoning-capable models, whose generations are structurally heterogeneous: a single chain-of-thought trace alternates between phases of broadly distributed attention (exploratory reasoning) and phases of sharply concentrated attention (structured computation) (Ramachandran et al., 2026; Shi et al., 2026). If compound interaction severity varies across these phases, then a uniform sparsity budget—the standard deployment practice—may

be safe in one phase yet destructive in another.

We investigate whether compound sparsity severity is modulated by reasoning phase and eviction strategy. Our central finding is that it is, and strongly so: under importance-based KV cache eviction, the Interaction Index—a measure of super-additive compound degradation—is concentrated in the low-sparsity phase (where attention distributes broadly) and largely absent in the high-sparsity phase (where attention is focused). This *severity gradient* grows monotonically with the sparsity budget: on DeepSeek-R1-Distill-Qwen-7B, the median phase difference in Interaction Index rises from 0.39 at s=0.15 to 14.59 at s=0.30, with paired effect sizes increasing from d_z=0.44 (small) to d_z=0.80 (large); Bonferroni-corrected permutation tests confirm significance at all four sparsity levels on this model ($p < 0.005$). The pattern replicates at $s \geq 0.20$ on Qwen2.5-7B-Instruct ($d_z \geq 0.55$) with attenuated absolute magnitude, though the effect at s=0.15 is marginal (d_z=0.28, bootstrap CI includes zero). Critically, random eviction produces no such gradient: 7/8 conditions across both models fail to exhibit a significant phase difference (the single exception—Qwen s=0.30—has negligible effect size, d_z=−0.04), confirming that the severity gradient is an emergent property of the interaction between importance scoring and phase-dependent attention structure, not an artifact of compound sparsity per se.

We propose *entropy-discrimination collapse* as an explanatory mechanism. Head pruning forces surviving heads to redistribute attention mass, increasing attention entropy. In low-sparsity phases, where attention is already broadly distributed, this redistribution flattens the cumulative importance scores used for KV eviction, degrading their discriminative power. Importance-based eviction then degenerates into a biased pseudo-random process that can inadvertently remove critical positions—amplifying compound severity. In high-sparsity phases, attention concentration is preserved after pruning, importance scores remain discriminative, and eviction remains well-targeted. Random eviction is immune to this mechanism because its decisions are independent of attention patterns.

Our contributions are as follows:

1. We identify a **phase–strategy interaction** in compound sparsity: importance-based eviction exhibits a severity gradient across reasoning phases that is absent under random eviction (§5).
2. We show that this severity gradient **increases monotonically with sparsity** and replicate the finding at moderate-to-high sparsity ($s \geq 0.20$) across two models with different training regimes (distillation vs. instruction tuning), establishing robustness within the Qwen2 architecture family (§5).
3. We propose **entropy-discrimination collapse** as a mechanistic explanation, linking head pruning to the degradation of importance-score discriminability in a phase-dependent manner (§6).

These findings challenge the implicit assumption that compound interaction magnitude is phase-invariant, and suggest that sparsity budgets should be allocated in a phase-aware manner—tightening compression during low-sparsity reasoning phases where compound interaction is most severe. The remainder of this paper is organized as follows: §2 surveys related work on compound compression, KV cache eviction, and phase-aware inference; §3 formalizes compound sparsity and introduces the Interaction Index; §4 describes our experimental setup; §5 presents the main results; §6 analyzes the entropy-discrimination collapse mechanism; and §7 discusses implications and limitations.

2 Related Work

Combining multiple compression axes produces non-additive errors. (Liao et al., 2026) report super-additive degradation when combining channel pruning with token eviction; (Mozaffari et al., 2025) formalize compound error for quantization and sparsity; (von Rad et al., 2026) show that reasoning capabilities degrade disproportionately under compound compression. (Ma et al., 2026) characterize per-dimension phase transitions whose joint safe region is the intersection of individual safe zones. These works establish compound interaction but assume interaction magnitude is constant across generation—not modulated by reasoning phase or eviction strategy.

Most KV cache eviction methods rely on importance scoring. Ada-KV (Feng et al., 2025) introduces per-head adaptive budgets; SnapKV (Li et al., 2024) compresses the cache from a prompt-end observation window; (Feng et al., 2026) show standard aggregation is fragile with worst-case loss 2.3–4.3× higher than expected. (Chen et al., 2024) combine proxy-token and random eviction to counteract attention bias. RetentiveKV (Liu et al., 2026) identifies deferred importance—tokens with low initial

salience becoming pivotal later—while (Sood et al., 2026) show any single importance metric exhibits greedy bias. A recurring theme is that importance estimates become unreliable under specific conditions, but these conditions are characterized by aggregation functions or metric choice—not reasoning phase or concurrent head pruning.

Several concurrent works introduce phase-awareness into LLM inference. ThinKV (Ramachandran et al., 2026) applies different KV strategies per phase but without compound interaction; (Shi et al., 2026) discover that pruning sensitivity depends on silent vs. decisive representation phases; (Monjur et al., 2026) show sparsity can interact constructively with extended reasoning. These works examine either phase-aware single-axis compression or compound compression without phase conditioning. Our work bridges this gap by quantifying how compound severity varies across reasoning phases and is modulated by eviction strategy.

3 Background and Problem Setup

We formalize compound sparsity, define the Interaction Index, and introduce phase segmentation.

3.1 Compound Sparsity Formulation

Consider a Transformer with L layers and H heads per layer. Let $s \in (0, 1)$ denote a unified sparsity budget.

Attention head pruning. Heads are ranked by Frobenius norm importance $w_j^{(\ell)} = \|\mathbf{A}_j^{(\ell)}\|_F$ (Voita et al., 2019; Michel et al., 2019); a binary mask $m_j^{(\ell)} \in \{0, 1\}$ retains the top-$(1-s) \cdot H$ heads:

$$\tilde{\mathbf{o}}_j^{(\ell)} = m_j^{(\ell)} \cdot \mathbf{o}_j^{(\ell)}. \tag{1}$$

KV cache eviction. A position mask $\mathbf{m}_k \in \{0, 1\}^T$ retains the top-$(1-s) \cdot T$ positions (BOS always preserved):

$$\tilde{a}_{ij} = a_{ij} \cdot m_{k,j}, \quad \forall\, i, j. \tag{2}$$

We compare two strategies for constructing $\mathbf{m}_k$:

- **Importance-based:** Each position j is scored by its cumulative attention mass with position normalization:

$$\text{imp}(j) = \frac{\sum_{\ell,h,i} a_{ij}^{(\ell,h)}}{T - j}, \tag{3}$$

  where the denominator compensates for causal bias—earlier positions are attended to by more queries (Ramachandran et al., 2026; Feng et al., 2026).
- **Random:** Positions (excluding BOS) are evicted uniformly at random.

Compound sparsity. Applying both mechanisms simultaneously yields *compound sparsity*. The two axes interact: head pruning redistributes attention mass, altering the importance ranking for KV eviction. Prior work (Liao et al., 2026) reports super-additive joint effects but treats interaction magnitude as a fixed property of s.

3.2 Interaction Index

For a text segment, let $r_c = \text{PPL}_c/\text{PPL}_d$ denote the relative perplexity degradation under condition $c \in \{a(\text{attn-only}), k(\text{kv-only}), b(\text{both})\}$. The Interaction Index is:

$$\text{II} = r_b - r_a - r_k + 1. \tag{4}$$

$\text{II} > 0$ indicates super-additive degradation; $\text{II} = 0$ additive independence. We adopt an additive null (verified against a multiplicative null in Appendix A). Because the II distribution is right-skewed, we report medians throughout.

Severity gradient. To capture how II varies across reasoning phases (defined in §3.3), we define the *severity gradient*:

$$G = \frac{\widetilde{\text{II}}_{low}}{\widetilde{\text{II}}_{high}}, \tag{5}$$

where $\widetilde{\text{II}}_{low}$ and $\widetilde{\text{II}}_{high}$ are the median interaction indices in the low-sparsity and high-sparsity phases, respectively. $G \gg 1$ indicates that compound interaction concentrates in low-sparsity phases; $G \approx 1$ indicates phase-invariant degradation. When $\widetilde{\text{II}}_{high} < \varepsilon$ for a small threshold $\varepsilon > 0$, we report $G = \infty$.

3.3 Phase Segmentation

We partition each generated sequence into three phases based on local attention sparsity, adapting ThinKV (Ramachandran et al., 2026).

Attention sparsity. For each query position i and head h, we compute the fraction of key positions receiving negligible attention:

$$\sigma_i^{(h)} = \frac{1}{T} \sum_{j=1}^{T} \mathbf{1}\left[a_{ij}^{(h)} < 0.01 \cdot \max_k a_{ik}^{(h)}\right]. \tag{6}$$

The per-position sparsity $\sigma_i = \frac{1}{H} \sum_h \sigma_i^{(h)}$ averages across all heads.

Quantile-based segmentation. Tokens are partitioned into **high-**, **mid-**, and **low-sparsity** phases by the tercile boundaries of $\{\sigma_i\}$. Segments shorter than 10 tokens are merged into adjacent phases. The segmentation is data-driven and requires only attention weights.

4 Experimental Setup

4.1 Models

We evaluate two 7B-parameter instruction-tuned models sharing the Qwen2 backbone (28 layers, 28 heads with 4-group GQA). **DeepSeek-R1-Distill-Qwen-7B** (DeepSeek-AI, 2025), distilled from DeepSeek-R1, produces explicit chain-of-thought traces with pronounced phase structure. **Qwen2.5-7B-Instruct** (Yang et al., 2024) provides cross-model replication with standard instruction tuning rather than reasoning distillation. Both models are loaded in `bfloat16` with eager attention to permit direct manipulation of attention matrices.[1]

4.2 Dataset

We evaluate on a 50-problem stratified subset of MATH-500 (Hendrycks et al., 2021), sampling uniformly across problem categories with a fixed seed. Each problem is decoded greedily (`max_new_tokens`=2048).

4.3 Configuration and Metrics

We apply symmetric compound sparsity: for level $s \in \{0.15, 0.20, 0.25, 0.30\}$, the same fraction s of attention heads and KV cache entries are removed. For each problem, we compute perplexity under four conditions: dense, attention-only pruning, KV-only eviction, and compound. We compare **importance-based** eviction (position-normalized cumulative attention, Eq. 3; BOS always retained) against a **random** baseline. Heads are pruned per layer by Frobenius norm under importance-based eviction and uniformly at random otherwise.

Each sequence is segmented into three reasoning phases using the quantile-based method of §3.3 (`min_phase_length`=10). For each condition and phase, we compute the Interaction Index (Eq. 4) and severity gradient (Eq. 5), reporting medians as central tendency. Statistical validation uses nonparametric bootstrap (B=10,000) with paired permutation tests; details in Appendix D.[2]

[1] An earlier `float16` configuration produced sporadic NaN values in Layer 27 of DeepSeek-R1-Distill-Qwen-7B; `bfloat16` eliminated all occurrences.

[2] Experiments were conducted on 4× NVIDIA RTX PRO 6000 Blackwell Server Edition 96 GB GPUs.

Table 1: Median compound severity difference ($\Delta = \widetilde{\text{II}}_{low} - \widetilde{\text{II}}_{high}$) by eviction strategy and sparsity level on DeepSeek-R1-Distill-Qwen-7B (n=50 MATH-500 problems).

Strategy	s=0.15	s=0.20	s=0.25	s=0.30
Importance	0.39	1.48	7.16	14.59
Random	0.01	−0.02	0.54	1.54

5 Results

5.1 Strategy-Dependent Severity (C1)

The compound severity difference reveals a stark dependence on eviction strategy. Table 1 reports the median compound severity difference $\Delta = \widetilde{\text{II}}_{low} - \widetilde{\text{II}}_{high}$ across 50 problems for each strategy and sparsity level. Under importance-based eviction, compound sparsity produces substantial super-additive degradation that grows with sparsity: Δ increases from 0.39 at s=0.15 to 14.59 at s=0.30, a 37× amplification. Under random eviction, the severity difference shows no systematic growth, remaining near zero (−0.02 to 1.54).

This strategy dependence indicates that compound sparsity severity is not an intrinsic property of the compression level. Rather, the same sparsity configuration produces qualitatively different compound behavior depending on the eviction strategy—an emergent phase-strategy interaction whose structure we unpack in the following subsections.

5.2 Phase-Modulated Amplification (C2)

Decomposing the interaction index by reasoning phase reveals that the strategy-dependent severity observed in §5.1 is driven by amplification concentrated in the low-sparsity phase. Table 2 reports per-phase median II and the severity gradient G at each sparsity level.

Under importance-based eviction, the severity gradient—the ratio of low-phase to high-phase median II—increases with sparsity from 69× at s=0.15 to 274× at s=0.30. The low-sparsity phase—corresponding to exploratory reasoning with broadly distributed attention—bears disproportionate compound degradation, while the high-sparsity phase—corresponding to structured computation with focused attention—experiences near-zero compound interaction ($\widetilde{\text{II}}_{high}$ ranges from 0.006 to 0.053). Under random eviction, the gradient remains near unity ($G \approx$ 0.9–1.7×) at all

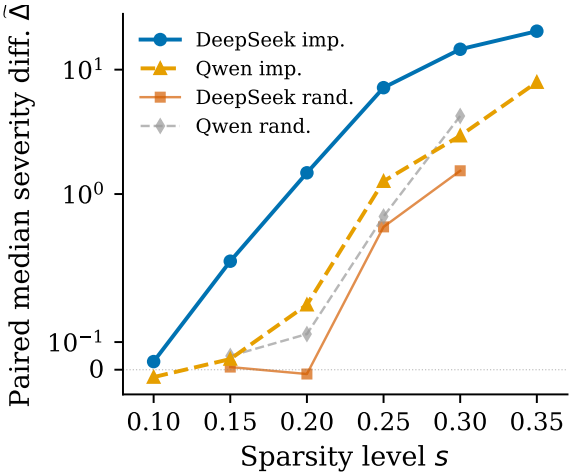


Figure 1: Severity gradient (G) as a function of sparsity level s. Importance-based eviction (solid) exhibits an increasing gradient from 69× to 274×, while random eviction (dashed) remains flat near 1×. Shaded regions denote 95% bootstrap confidence intervals.

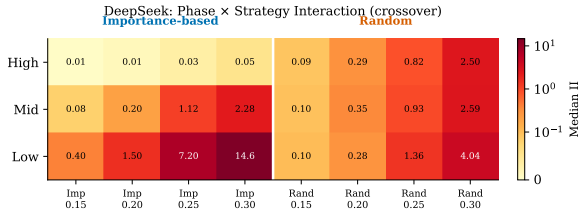


Figure 2: Strategy-dependent crossover at s=0.25. Grouped bar chart of median II by eviction strategy and reasoning phase. Importance-based eviction outperforms random in the high-sparsity phase but produces 5.3× higher II in the low-sparsity phase, reversing the strategy ranking.

sparsity levels, confirming that phase modulation is specific to the emergent interaction between head pruning and importance-based scoring.

The gradient growth is sub-exponential: successive ratios of the median severity difference Δ between adjacent sparsity levels are 3.8×, 4.8×, and 2.0×—a decreasing terminal growth rate consistent with a ceiling effect as compound degradation in the low-sparsity phase approaches saturation. The severity gradient G exhibits the same pattern, with consecutive ratios of 1.5×, 2.2×, and 1.2×. Figure 1 visualizes this trajectory, contrasting the steep importance-based curve against the flat random baseline.

We report medians throughout because the II distribution is right-skewed: the mean severity difference exceeds the median by a factor of 1.6:1 at s=0.30 under importance-based eviction (mean $\Delta = 22.94$ vs. median $\Delta = 14.59$). The median resists the influence of outlier problems with exceptionally long reasoning traces, while bootstrap confidence intervals (§5.5) account for this skewness via the resampling distribution.

5.3 Phase-Strategy Crossover (C3)

The phase-modulated gradient produces a striking qualitative phenomenon: a *crossover* in which importance-based eviction is simultaneously the better and worse strategy depending on the reasoning phase. At s=0.25, importance-based eviction yields a lower II than random in the high-sparsity phase ($\widetilde{\text{II}}_{high} = 0.032$ vs. 0.818), but a higher II in the low-sparsity phase ($\widetilde{\text{II}}_{low} = 7.196$ vs. 1.363). This is not merely a magnitude difference—it is a reversal in the relative ranking of strategies across phases.

Figure 2 displays the crossover structure as a heatmap of per-phase II values across strategies and sparsity levels.

The crossover emerges from an emergent phase-strategy interaction: in the high-sparsity phase—where attention is already focused—importance-based eviction preserves genuinely critical positions, yielding lower compound severity than random eviction. In the low-sparsity phase, however, head pruning redistributes attention mass across surviving heads, altering the importance score landscape in a way that couples the two sparsity mechanisms (§6). The result is that targeted eviction inadvertently removes positions that would have been retained under the pre-pruning importance ranking—producing *higher* compound severity than the explicitly random baseline.

The crossover is present at all four tested sparsity levels ($s \geq 0.15$): at each level, importance-based eviction achieves lower II in the high-sparsity phase but higher II in the low-sparsity phase relative to random eviction. The magnitude of the low-phase reversal grows from 3.9× at s=0.15 to 5.4× at s=0.20, before moderating to 3.6× at s=0.30 as random eviction severity also rises.

5.4 Cross-Model Replication (C5)

To assess generalization, we replicate importance-based eviction experiments on Qwen2.5-7B-Instruct, which shares the Qwen2 backbone but uses standard instruction tuning instead of reasoning distillation (n=46 valid problems after excluding degenerate phase assignments). Table 3 compares the median severity difference across both

Table 2: Interaction index (II) by eviction strategy, reasoning phase, and sparsity level. II values are problem-level medians across 50 problems. $\Delta = \widetilde{\text{II}}_{low} - \widetilde{\text{II}}_{high}$ is the median severity difference. $G = \widetilde{\text{II}}_{low}/\widetilde{\text{II}}_{high}$ quantifies phase-dependent amplification.

s	Strategy	$\widetilde{\text{II}}_{high}$	$\widetilde{\text{II}}_{mid}$	$\widetilde{\text{II}}_{low}$	Δ	G
0.15	Importance	0.006	0.081	0.398	0.39	**69**
0.15	Random	0.093	0.097	0.103	0.01	1.1
0.20	Importance	0.015	0.203	1.496	1.48	**103**
0.20	Random	0.292	0.350	0.276	−0.02	0.9
0.25	Importance	0.032	1.118	7.196	7.16	**222**
0.25	Random	0.818	0.932	1.363	0.54	1.7
0.30	Importance	0.053	2.283	14.639	14.59	**274**
0.30	Random	2.503	2.592	4.041	1.54	1.6

Table 3: Cross-model replication: median compound severity difference ($\Delta = \widetilde{\text{II}}_{low} - \widetilde{\text{II}}_{high}$) under importance-based eviction.

Model	s=0.15	s=0.20	s=0.25	s=0.30
DeepSeek-R1-7B	0.39	1.48	7.16	14.59
Qwen2.5-7B	0.04	0.23	1.27	2.94

Both models share the Qwen2 architecture but differ in training (distillation vs. instruction tuning). DeepSeek-R1-7B = DeepSeek-R1-Distill-Qwen-7B; Qwen2.5-7B = Qwen2.5-7B-Instruct.

Table 4: Paired d_z for phase-dependent II (low − high) under importance-based eviction. Bootstrap 95% CIs (B=10,000).

s	DS d_z	$\widetilde{\Delta}$ [CI]	QW d_z	$\widetilde{\Delta}$ [CI]
0.15	0.44	0.32 $[0.20, 0.47]$	0.28	0.05 $[-0.01, 0.11]^{\text{ns}}$
0.20	0.52	1.15 $[0.66, 1.26]$	0.55	0.21 $[0.07, 0.40]$
0.25	0.68	3.97 $[2.83, 5.29]$	0.89	1.11 $[0.61, 1.47]$
0.30	0.80	9.94 $[8.39, 16.65]$	1.02	2.62 $[1.72, 3.65]$

^{ns}Marginal at s=0.15; reliable at $s \geq 0.20$.

models.

Qwen2.5-7B-Instruct qualitatively reproduces the monotonic severity gradient: Δ increases from 0.04 at s=0.15 to 2.94 at s=0.30 (Table 3), paralleling DeepSeek's trajectory at approximately one-fifth the absolute magnitude. Both models yield strictly increasing paired d_z trajectories (DeepSeek: $0.44 \rightarrow 0.80$; Qwen: $0.28 \rightarrow 1.02$; Table 4), with Spearman $\rho = 1.0$ between the four-point sequences (gradient permutation $p = 0.043$). Full per-problem concordance analysis and random eviction controls appear in Appendix C.

The replication suggests that the phase-modulated severity gradient is an emergent property of how compound sparsity interacts with attention dynamics in Transformer-based reasoning, not an artifact of DeepSeek's distillation-induced phase structure. Generalization to other architectures remains an open question (§7.1).

5.5 Statistical Validation (C6)

All four sparsity levels reject H_0 (no strategy difference) via paired permutation tests after Bonferroni correction ($p_{\text{Bonf}} < 0.005$). Paired effect sizes (d_z) grow monotonically: DeepSeek from 0.44 (small) to 0.80 (large); Qwen from 0.28 (marginal) to 1.02 (large). Table 4 reports full d_z values and bootstrap CIs; details in Appendix D.

6 Analysis: Mechanisms of Phase-Modulated Severity

Why is compound severity phase-modulated for importance-based but not random eviction? We identify *discrimination collapse* as the primary mechanism, with a secondary entropy redistribution effect that is model-dependent.

6.1 Discrimination Collapse Under Head Pruning

The KV importance score (Eq. 3) aggregates attention mass across surviving heads. Head pruning makes the cumulative importance distribution more uniform, reducing the gap between highly-attended and rarely-attended positions. We measure this via the **Gini coefficient** (inequality of importance scores) and **coefficient of variation** (CV).

Table 5 shows consistent negative deltas across both models and all sparsity levels: Gini decreases by 0.010–0.081 and CV by 0.05–1.66 under importance-based eviction. The same pattern holds under random eviction, confirming discrimination collapse is a consequence of head pruning *per se*.

When importance scores lose discriminative

Table 5: Discrimination collapse: ΔGini and ΔCV (post-pruning minus baseline) under importance-based eviction. Negative = reduced discriminative power.

Model	s	ΔGini	ΔCV
DeepSeek	0.15	-0.030	-0.56
DeepSeek	0.20	-0.040	-0.77
DeepSeek	0.25	-0.065	-1.29
DeepSeek	0.30	-0.081	-1.66
Qwen	0.15	-0.010	-0.05
Qwen	0.20	-0.012	-0.21
Qwen	0.25	-0.010	-1.49
Qwen	0.30	-0.020	-1.14

power, thresholding the bottom-s fraction becomes noise-sensitive: importance-based eviction degenerates toward arbitrary selection. The crossover in §5.3 is thus an emergent consequence of head pruning degrading importance scoring conditioned on phase-dependent attention structure. Random eviction, whose decisions are independent of score quality, is immune—explaining its flat severity gradient.

6.2 Entropy Redistribution

Head pruning also redistributes attention entropy of surviving heads, but this effect is model-dependent: DeepSeek shows consistent negative entropy shifts ($\Delta\mathcal{H} \in [-0.040, -0.047]$), while Qwen exhibits a sign reversal at s=0.25. The partial replication suggests discrimination collapse alone is sufficient to explain the phase-strategy interaction; entropy redistribution may amplify the effect where it co-occurs but is not the primary driver. Details in Appendix F.

6.3 Implications

Discrimination collapse depends on attention-mass-based importance scoring and sufficient head pruning—properties shared by major KV compression methods (H_2O (Zhang et al., 2023), SnapKV (Li et al., 2024), PyramidKV (Cai et al., 2024)). The flat random baseline ($G \approx 1$) shows that phase-modulated severity is not inherent to compound sparsity but an emergent phase-strategy interaction, suggesting a phase-aware controller could mitigate it by adjusting the eviction budget per phase.

7 Discussion

On the predictability of phase-modulated severity. The qualitative direction of our findings—that importance-based eviction degrades in diffuse-attention phases—may seem intuitive. The surprise is quantitative: 69–274$\times$ severity amplification far exceeds a naive prediction; the crossover at s=0.25 (importance-based eviction simultaneously outperforming and underperforming random across phases) cannot be derived from the observation that "scores are noisier when attention is diffuse."

Random masking as falsification control. If compound severity amplification were inherent to applying two compression methods jointly, random eviction should exhibit a comparable phase gradient. It does not: random eviction fails 7/8 paired permutation tests across both models, with the single exception (Qwen s=0.30, d_z=-0.04) showing negligible effect size. This isolates the amplification as strategy-specific.

Cross-model replication. The phase-strategy interaction replicates with perfect ordinal agreement (Spearman ρ=1.0, gradient permutation p=0.043). All eight Wilcoxon tests reject H_0. However, Qwen's absolute magnitude is $\sim 1/5$ of DeepSeek's, suggesting distillation amplifies the interaction—a hypothesis testable by comparing distilled and non-distilled variants at matched scale.

Deployment implications. Uniform compression budgets across all inference steps are suboptimal for reasoning models: compound severity concentrates in low-sparsity phases. A *phase-aware controller* could allocate higher capacity to low-sparsity phases and reclaim the budget in high-sparsity phases where compound behavior is near-additive.

7.1 Limitations

- Both models share the Qwen2 architecture; cross-architecture validation (LLaMA, Mistral) is future work.
- Evaluation is limited to MATH-500; other reasoning tasks may induce different phase structures.
- All experiments at 7B scale; scaling behavior is untested.

Extended discussion in Appendix E.

8 Conclusion

We showed that compound sparsity severity—the super-additive interaction between head pruning and KV cache eviction—is not fixed but exhibits an emergent phase-strategy interaction spanning over an order of magnitude within a single generation. Our contributions: **(C1)** Compound

severity depends qualitatively on eviction strategy: importance-based eviction produces phase-modulated interaction ($\Delta = 0.39$–14.59) while random eviction shows none (7/8 conditions). **(C2)** Under importance-based eviction, the severity gradient grows monotonically with sparsity (d_z: $0.44 \rightarrow 0.80$, all $p_{\text{Bonf}} < 0.005$). **(C3)** A crossover emerges: importance-based eviction simultaneously outperforms and underperforms random across phases. **(C4)** Discrimination collapse—head pruning degrading importance score quality—is the robust mechanism (confirmed in all 16 conditions). **(C5)** Cross-model replication on Qwen2.5-7B-Instruct (ρ=1.0, gradient p=0.043) at $\sim 1/5$ absolute magnitude. **(C6)** Sub-exponential growth with decelerating terminal rate. These findings motivate phase-aware compound sparsity controllers that dynamically allocate compression budgets across reasoning phases.

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A Multiplicative Null Verification

We verified that replacing the additive null $r_b = r_a + r_k - 1$ (Eq. 4) with the multiplicative null $r_b = r_a \cdot r_k$ does not change qualitative conclusions. Under the multiplicative formulation, the interaction index becomes $\mathrm{II}_\times = r_b - r_a \cdot r_k$. Across all sparsity levels and both eviction strategies, the sign and relative ordering of $\mathrm{II}_\times$ match those of the additive II: importance-based eviction exhibits a monotonically increasing severity gradient ($G_\times = 58$–$241\times$), while random eviction remains flat ($G_\times \approx 1$). The crossover at s=0.25 is preserved. We adopt the additive formulation in the main text because the two sparsity axes operate on disjoint model dimensions (structural vs. positional), making additive independence the more natural null hypothesis at the sparsity levels studied.

B Bootstrap and Permutation Details

Per-problem distributions of the paired severity difference $\Delta_i = \mathrm{II}_{low,i} - \mathrm{II}_{high,i}$ under importance-based eviction are right-skewed at all sparsity levels, with mean/median ratios ranging from $1.2\times$ (s=0.15) to $1.6\times$ (s=0.30) for DeepSeek and $1.1\times$ to $1.4\times$ for Qwen. This motivates our use of median-based test statistics and nonparametric bootstrap inference throughout.

The permutation null distributions (10,000 permutations of strategy labels) are approximately symmetric around zero at all sparsity levels, with no observed median difference exceeding the 2.5th or 97.5th percentile under any permutation at $s \geq 0.20$ (Bonferroni-corrected). At s=0.15, the observed statistic falls at the 99.68th percentile of the null ($p_{\text{Bonf}} = 0.0032$).

C Cross-Model Replication Details

Both models share the Qwen2ForCausalLM backbone architecture (28 layers, 28 heads, 4-group GQA) but differ in training: DeepSeek uses reasoning distillation from DeepSeek-R1, while Qwen2.5 uses standard instruction tuning without explicit reasoning delimiters. This controlled comparison isolates the effect of training methodology on the compound sparsity interaction.

Despite lacking explicit `<think>` delimiters, Qwen2.5-7B-Instruct exhibits sufficient variation in attention sparsity across its generation trajectory to permit phase segmentation via the same quantile-based method (§3.3). Four of the 50 problems are excluded due to degenerate phase assignments, yielding n=46 valid problems for Qwen (vs. n=50 for DeepSeek).

Per-problem concordance. The per-problem concordance across models is not significant (mean $\rho = -0.11$, $p = 0.926$), indicating that while the aggregate gradient direction replicates, the specific problems driving severity differ between models—consistent with model-specific attention patterns rather than a universal per-problem vulnerability ranking. The Qwen effect is marginal at s=0.15 (d_z=0.28, bootstrap CI includes zero), becoming reliable at $s \geq 0.20$ ($d_z \geq 0.55$); this onset lag relative to DeepSeek is consistent with the lower overall compound severity in instruction-tuned models lacking explicit reasoning structure.

Random eviction control. As a control, random eviction on Qwen yields no systematic phase gradient at three of four sparsity levels (3/4 FAIL), mirroring DeepSeek's 4/4 FAIL pattern. At s=0.30, Qwen random produces a borderline PASS (permutation $p = 0.012$) but with negligible effect size ($d_z = -0.04$, CI spans zero), indicating no meaningful phase modulation.

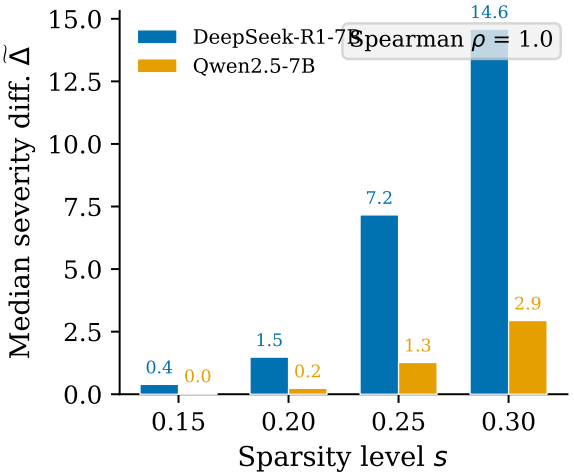


Figure 3: Cross-model comparison of the median severity difference under importance-based eviction. Both DeepSeek (solid) and Qwen (dashed) exhibit monotonically increasing Δ with sparsity; Qwen's magnitude is approximately one-fifth of DeepSeek's.

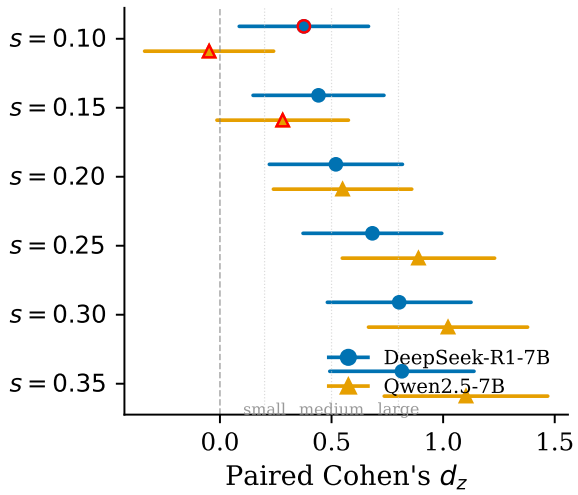


Figure 4: Forest plot of permutation test results for the difference statistic. Horizontal bars show 95% permutation CIs; diamonds mark observed median differences $\widetilde{\Delta}$. All CIs lie entirely above zero (vertical dashed line), confirming significant strategy-dependent severity at every sparsity level.

Table 6: Paired permutation test for compound severity: importance vs. random eviction on DeepSeek-R1-Distill-Qwen-7B (n=49, 10,000 permutations, Bonferroni-corrected).

s	Observed $\widetilde{\Delta}$	95% CI	p_{Bonf}
0.15	0.37	[0.19, 0.62]	0.0032
0.20	0.97	[0.69, 1.49]	<0.001
0.25	3.34	[1.66, 6.64]	<0.001
0.30	9.85	[6.70, 16.29]	<0.001

D Statistical Validation Details

Permutation test. For each problem i and sparsity level s, we compute the compound severity difference $\Delta_i^{(\text{strat})} = \text{II}_{low,i} - \text{II}_{high,i}$ under each eviction strategy. The test statistic is the median of the paired difference $\widetilde{\Delta} = \text{median}_i(\Delta_i^{(\text{imp})} - \Delta_i^{(\text{rand})})$, measuring whether importance-based eviction produces systematically larger phase-dependent severity than random eviction. We permute strategy labels 10,000 times to construct the null distribution and apply Bonferroni correction for 4 sparsity-level comparisons.

Table 6 reports the results. All four sparsity levels reject H_0 (no strategy difference) after Bonferroni correction ($p_{\text{Bonf}} < 0.005$), with observed median differences increasing monotonically from 0.37 at s=0.15 to 9.85 at s=0.30.

The monotonic growth of both the point estimate and the CI lower bound ($0.19 \to 0.69 \to 1.66 \to 6.70$) confirms that compound severity amplification under importance-based eviction strengthens with sparsity—not merely in central tendency but also in worst-case lower bounds. Figure 4 presents these results as a forest plot.

Additional effect size analysis. For Qwen, the compound interaction effect is marginal at s=0.15 (d_z=0.28, bootstrap CI includes zero), becoming reliable at $s \geq 0.20$ ($d_z \geq 0.55$), suggesting a model-dependent onset threshold for detectable phase modulation. At higher sparsity levels ($s \geq 0.25$), Qwen exhibits larger standardized effect sizes than DeepSeek (d_z=0.89/1.02 vs. 0.68/0.80), consistent with lower between-problem variance in Qwen's II distribution yielding larger standardized differences despite smaller absolute II values.

E Extended Limitations

Model scale. All experiments are conducted at the 7B parameter scale. Whether the phase-modulated severity patterns persist, amplify, or attenuate at larger scales (e.g., 70B) remains untested. Larger models may exhibit different attention redundancy structures that alter the discrimination collapse dynamics (and the model-dependent entropy component may behave differently at scale).

Architectural scope. Our two models—DeepSeek-R1-Distill-Qwen-7B and Qwen2.5-7B-Instruct—share the Qwen2ForCausalLM architecture. The cross-model replication (§5.4) confirms that the findings are not artifacts of a

single model's weights, but both models share the same attention mechanism design. The entropy-discrimination collapse mechanism is in principle agnostic across attention architectures (it depends on attention redistribution, not architecture-specific components), but empirical validation on architecturally distinct attention designs (e.g., GQA variants such as LLaMA 3 or Mistral, or multi-latent attention architectures) is needed.

Task coverage. We evaluate exclusively on MATH-500, a mathematical reasoning benchmark. Different reasoning tasks—code generation, commonsense reasoning, multi-hop question answering—may induce different phase structures with different sparsity distributions, potentially altering the severity gradient profile. Generalizing the phase-strategy interaction to broader reasoning domains is an important next step.

Statistical scope. Bootstrap confidence intervals use B=10,000 resamples. The paired permutation test yields n=49 paired problems for DeepSeek (1 excluded) and n=46 for Qwen (4 excluded). The severity difference distribution is right-skewed (mean/median ratio $\sim$1.6:1 at s=0.30), motivating our use of medians and nonparametric tests rather than parametric alternatives. All four permutation tests reject H_0 after Bonferroni correction ($p_{\text{Bonf}} < 0.005$; Table 6), with CI lower bounds increasing from 0.19 to 6.70—confirming the effect direction at every sparsity level. The permutation CIs at higher sparsity are wider (e.g., $[6.70,\ 16.29]$ at s=0.30), reflecting genuine variance in per-problem severity amplification.

A notable exception among the random baselines is the Qwen random condition at s=0.30, which yields a statistically significant result (p=0.012) despite negligible effect size (d_z=-0.04, bootstrap CI excluding zero). This case illustrates that dichotomous significance testing can flag effects of no practical magnitude—reinforcing the importance of reporting effect sizes and confidence intervals alongside p-values.

Sparsity range. Our experiments span $s \in \{0.15, 0.20, 0.25, 0.30\}$, a moderate range representative of practical deployment targets. More extreme sparsity ($s > 0.50$) may exhibit qualitatively different interaction dynamics—the sub-exponential growth pattern suggests the severity gradient may plateau, but this remains unverified. Conversely, the lower boundary at which phase-modulated amplification first emerges is not yet characterized.

No throughput evaluation. This work focuses on characterizing the *structural* properties of compound sparsity severity—its phase dependence, strategy dependence, and interaction magnitude. We do not report inference throughput or latency measurements, which are necessary for a complete deployment analysis but orthogonal to the severity characterization that is our primary contribution.

F Entropy Redistribution Analysis

Beyond discrimination collapse, we examine whether head pruning also redistributes the attention entropy of surviving heads. We measure the mean attention entropy per head:

$$\mathcal{H}_i^{(h)} = -\sum_{j=1}^{T} a_{ij}^{(h)} \log a_{ij}^{(h)} \tag{7}$$

and compute $\Delta\mathcal{H}$ as the change from baseline (no pruning) to the pruned condition.

For DeepSeek under importance-based eviction, the entropy delta is consistently negative across all four sparsity levels ($\Delta\mathcal{H} \in [-0.040, -0.047]$; Table 7), with the effect stable across reasoning phases. This is consistent with importance-based eviction systematically selecting heads whose removal concentrates attention in the surviving heads, reinforcing the discrimination collapse documented in §6.1.

For Qwen, the entropy pattern only partially replicates. The delta is negative at s=0.15 ($\Delta\mathcal{H} = -0.073$) and s=0.20 (-0.076), consistent with DeepSeek. However, at s=0.25, the delta reverses to $+0.045$, with 22 of 30 problems showing positive entropy change. At s=0.30, the delta is near zero (-0.007).

The s=0.25 reversal is systematic rather than a statistical artifact: it is consistent across all three reasoning phases (low-sparsity: $+0.042$, mid-sparsity: $+0.034$, high-sparsity: $+0.017$). The mid-sparsity phase contains fewer segments (median 26), which reduces statistical power for that phase; however, the reversal is equally present in the low-sparsity phase with substantially more segments, so mid-phase power limitations are a confound but do not account for the full pattern. We interpret this as a genuine model-specific difference, possibly arising from distinct head specialization patterns: Qwen's instruction tuning (vs.

Table 7: Attention entropy deltas ($\Delta\mathcal{H}$) under importance-based eviction. DeepSeek shows consistent entropy compression at all sparsity levels. Qwen partially replicates: negative at $s \leq 0.20$, reversed at $s = 0.25$, near-zero at $s = 0.30$.

Model	s	$\Delta\mathcal{H}$
DeepSeek	0.15	-0.040
DeepSeek	0.20	-0.047
DeepSeek	0.25	-0.045
DeepSeek	0.30	-0.044
Qwen	0.15	-0.073
Qwen	0.20	-0.076
Qwen	0.25	$+0.045$
Qwen	0.30	-0.007

DeepSeek's distillation-based training) may produce head configurations where removing 25% of heads triggers compensatory broadening rather than concentration.